

Customer Behaviour Analysis

About the Dataset

- **Type:** Retail / E-commerce Dataset
- **Nature:** Structured (Tabular)
- **Level:** Customer Transaction Level
- **Data Mix:**
 - Numerical → Age, Purchase Amount, Previous Purchases
 - Categorical → Category, Payment Method, Season

This dataset is used for **customer behavior analysis + business decision-making**

Domain Background

In the **e-commerce industry**, companies track every customer interaction:

- What customers buy
- How often do they buy
- How much do they spend
- Whether they respond to discounts

Why This Data Is Important

This data helps businesses:

- Identify **high-value customers**
- Understand **purchase patterns**
- Optimize **discount strategies**
- Improve **marketing campaigns**
- Increase **customer lifetime value (CLV)**

Business Problem

How can a retail company understand customer purchasing behavior to increase revenue, improve customer retention, and optimize marketing strategies?

Breaking Down the Problem

Low Customer Retention

- Customers are not purchasing frequently

Need to understand:

- Who are frequent buyers?
- Who are one-time buyers?

Ineffective Marketing Campaigns

- Discounts and promo codes may not be effective

Questions:

- Do discounts increase purchases?
- Which customers respond to promo codes?

Product Demand Understanding

- Which category performs best?
- Which products generate the highest revenue?

Payment Behavior Insights

- Which payment methods are preferred?
- Are digital payments increasing?

Seasonal Demand Variation

- Does season impact sales?
- Which season drives maximum purchases?

Customer Segmentation Problem

Need to identify:

- High-value customers
- Frequent buyers
- Discount-sensitive customers

Data Dictionary

Column Name	Description
Customer ID	Unique identifier for each customer
Age	Age of the customer
Gender	Gender of the customer (Male / Female / Other)
Item Purchased	Specific product purchased
Category	Product category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent by the customer
Location	Customer's geographic location
Size	Product size (S, M, L, etc.)
Color	Product color
Season	Season of purchase (Winter, Summer, etc.)
Review Rating	Customer rating for the product
Subscription Status	Whether customer is subscribed (Yes / No), Subscription Program
Shipping Type	Type of shipping selected
Discount Applied	Whether discount was applied
Previous Purchases	Number of past purchases
Payment Method	Mode of payment
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

EDA – Exploratory Data Analysis using Python

Data Loading

What you did:

- Imported **pandas**
- Loaded dataset using:

```
#importing the dataframe/dataset  
df = pd.read_csv('customer_shopping_behavior.csv')
```

Why did you do it:

- To bring raw data into a structured DataFrame
- Enables further analysis and transformations

Initial Data Exploration

What you did:

- Viewed data using:
 - df.head()
 - df.tail()
- Checked dataset summary:
 - df.describe(include='all')
 - df.info()
- Checked missing values:
 - df.isnull().sum()

Why did you do it?

- To understand: Data structure, Data types, Missing values
- This step helps identify **data quality issues early**

Data Consistency Check (Category Correction)

What you did:

- Checked unique values in:
 - Item Purchased
 - Category
- Found inconsistencies between **Item Purchased** and **Category**
- Created a **mapping dictionary**
- Updated category using it

Why did you do it?

- Categories were inconsistent or incorrect
- Ensures:
 - Accurate grouping
 - Correct business insights

This is a **critical real-world data cleaning step**

Handling Missing Values (Size Column)

What you did:

Handled missing values based on category:

- Electronics → "Not Applicable"
- Accessories → "Not Applicable"
- Footwear → "Free Size"
- Clothing → Filled with **mode**

Why did you do it?

- Different categories require **different business logic**
- Example:
 - Electronics don't have size → Not Applicable
 - Clothing → size is important → use mode

This shows **domain understanding**

Handling Missing Values (Review Rating)

What you did:

- Checked: Mean, median, mode
- Filled missing values using product-level mean
- Rounded values

Why did you do it?

- Instead of the global average, the **product-level mean is used**
- This keeps ratings **realistic and meaningful**

This is an **advanced imputation technique**

Handling Missing Values (Purchase Amount)

What you did: Replaced with the mean value.

Why did you do it?

- Maintains **product-level pricing consistency**
- Avoids distortion in revenue analysis

Handling Missing Values (Previous Purchases)

What you did: Replaced with the 0 (zero) value.

Why did you do it?

- Missing values imply **no previous purchase**
- Important for:
 - Customer segmentation
 - Loyalty analysis

Duplicate Handling

What you did:

- Checked duplicates
- Identified duplicates using Customer ID
- Removed duplicates

Why did you do it?

- Ensures: Each customer appears only once
- No double-counting in analysis

Critical for **accurate reporting**

Column Standardization

What you did:

- Converted column names:

```
df.columns = df.columns.str.replace(" ", "_")  
df.columns = df.columns.str.lower()
```

Renaming the Headers

```
df.columns = df.columns.str.replace(" ", "_")  
df.columns = df.columns.str.lower()
```

Why did you do it?

- Makes column names:
 - Clean
 - SQL-friendly
 - Easy to use in Power BI

Data Export to SQL Server

What you did:

- Installed required libraries:

```
pip install pyodbc sqlalchemy
```

- Connected to SQL Server
- Exported data

Why did you do it?

- To move cleaned data into **SQL for further analysis**

Enables:

- Query-based insights
- Integration with BI tools

Data Analysis using SQL Server

1. Which category generates the highest revenue?

Category	Highest Revenue
Electronics	378,785.61
Accessories	194,866.87
Clothing	183,188.39
Footwear	111,236.20
Outerwear	18,524.00

2. Are discounts actually increasing purchase value?

Discount Applied	Total Revenue	Avg Purchase
No	447,484.74	167.16
Yes	439,116.34	202.73

3. What is the total revenue generated by male and female customers?

Gender	Highest Revenue
Male	358,295.55
Female	304,964.90
Other	223,340.64

4. Which customer used a discount but still spent more than the average purchase amount?

Customer ID	Purchase Amount	Discount Applied
4568	276.34	Yes
4256	734.48	Yes
4289	1,282.44	Yes
4576	831.34	Yes
4708	735.77	Yes
4607	557.24	Yes
4845	692.11	Yes
4558	208.02	Yes
4492	805.50	Yes
4292	805.50	Yes

5. Which are the top / bottom 5 products with the highest average review rating?

Top 5 Products by Avg Rating:

Item Purchased	Avg Ratings
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.79

Bottom 5 Products by Avg Rating:

Item Purchased	Avg Ratings
Phone	2.94
Headphones	3.08
Bag	3.17
Laptop	3.19
Shirt	3.42

6. Average purchase by shipping type: Standard, Express, Free, Store Pickup, 2-Day Shipping, Next Day Air

Shipping Type	Orders Placed	Avg Purchase	Revenue
Standard	1,136	322.39	366,232.05
Express	1,107	329.35	364,588.03
Free Shipping	675	60.41	40,777.00
Store Pickup	650	59.89	38,931.00
2-Day Shipping	627	60.73	38,080.00
Next Day Air	648	58.63	37,993.00

7. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

Subscription Status	Users	Avg Revenue	Total Revenue
No	3,334	152.33	507,858.75
Yes	1,509	250.99	378,742.34

8. Top 5 products with highest discount usage %

Item Purchased	Total Times Sold	Times Sold with Discount	Discount %
Laptop	143	74	51.75
Phone	171	88	51.46
Headphones	166	85	51.20
Hat	154	77	50.00
Sneakers	145	72	49.66

9. Segment customers into new, returning, and loyal based on their total number of previous purchases, and show the count of each segment.

Customer Segment	Customer Count
Loyal Customers	3,034
Returning Customer	1,332
New Customer	477

10. What are the top 3 most purchased products within each category?

Category	Item Purchased	Most Purchased	Rank
Accessories	Bag	164	2
Accessories	Sunglasses	161	3
Accessories	Belt	161	3
Clothing	Shirt	315	1
Clothing	Pants	171	2
Clothing	Blouse	171	2
Electronics	Phone	171	1
Electronics	Headphones	166	2
Electronics	Laptop	143	3
Footwear	Shoes	303	1
Footwear	Sandals	160	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

11. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

Customer Type	Subscription Status	Customer Count	Percent
Normal Buyers	No	612	64.02%
Normal Buyers	Yes	344	35.98%
Repeat Buyers	No	2,722	70.03%
Repeat Buyers	Yes	1,165	29.97%

12. What is the revenue contribution of each age group?

Age Group	Total Revenue
51+	354,427.43
35–50	238,151.41
26–35	163,048.71
18–25	130,973.53



CUSTOMER BEHAVIOUR ANALYSIS

Category
All

Gender
Female Male Other

Discount Applied
No Yes

Subscription Status
No Yes



5K
Total Customers

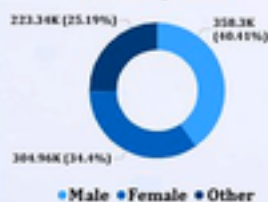
29
Unique Items

3.63
Average Review

886.60K
Total Spend

183.07
Average Spend

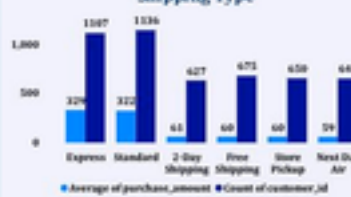
Revenue by Gender



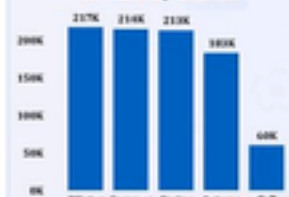
Revenue by Category



Shipping Type



Revenue by Seasons



Average Revenue by Location



Revenue by Age Distribution



Revenue by Payment Method





CUSTOMER BEHAVIOUR ANALYSIS

Category
All

Gender

Female

Male

Other

Discount Applied

No

Yes

Subscription Status

No

Yes



5K
Total Customers



29
Unique Items



3.63
Average Review



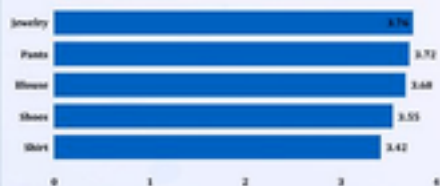
886.60K
Total Spend



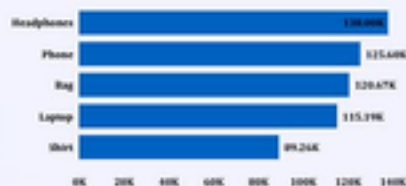
183.07
Average Spend



Top 5 Items by Ratings



Top 5 Items by Revenue



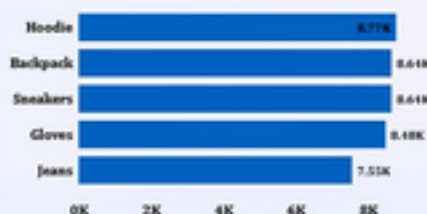
Top 5 Location by Revenue



Bottom 5 Items by Ratings



Bottom 5 Items by Revenue



Bottom 5 Location by Revenue

